

Anurag Aladhahalli Sampath

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Education

SJCE Mysuru, IN, BE Electronics & Communication Engineering

2015 – 2019

- GPA: 9.71 (Rank 3/120)

Professional Experience

Lead Engineer, Capital One – London, UK

May 2026 – Present

Software Engineer II, Amazon – London, UK

Oct 2023 – Apr 2026

- Architected and led the development of a closed loop network monitoring system for 700K networking devices with ~11,000+Tps load resulting in reduction of production issues by 97%.
- Led a team of 3 engineers to build a CI/CD pipeline to streamline the production cycle for ML models.
- Built the Frontend and Backend components and APIs for the Production Assessment, Localization and VFX features of GenAI based script breakdown tool, saving 80k+ hours and processed ~500 movie scripts/day.
- Migrated REST APIs to GraphQL and integrated with Studios DataMesh Infrastructure to decentralize data for Horizon Analyze and optimized client performance by ~20% and reducing data load time from 5s to 1s.

Software Engineer I, Amazon – London, UK

Apr 2022 – Sep 2023

- Designed a scalable event-sourcing system using microservices architecture that integrated data from multiple third-party sources and uses ML models to assess the impact of special events on Prime Video.
- Developed a distributed systems architecture to onboard 100 metrics/PV device to a monitoring system, eliminating manual SDE effort and 6 weeks of onboarding time.

Verification Engineer, Arm – Bengaluru, IN

Oct 2020 – Apr 2022

- Explored novel bug detection techniques for CPUs, implemented checkers that uncovered bugs dating back several years, and spearheaded the single-core random testing.

Graduate Engineer, Arm – Bengaluru, IN

Jul 2019 – Sep 2020

- Played a pivotal role in the successful delivery of the Functional Safety CPU certification and worked on the directed verification program for A-Class CPUs.

Software Engineer Intern, LogMeIn – Bengaluru, IN

Jan 2019 – Jun 2019

- Contributed to the Boston Innovation Labs team by developing three POCs - Single Sign-On, Tweets Ingestion Pipeline, and GoToWebinar Breakout Sessions - that were successfully integrated into LogMeIn's product lineup.

Personal Projects

Bonsai 🌳

- Self-tuning spatial index library in Rust that automatically migrates between R-tree, KD-tree, Grid, and Quadtree at runtime based on live data.

Caravault 🚐

- Engineered an air-gapped peer-to-peer sync system achieving quorum-based conflict resolution across offline drives using version vectors and Merkle tree diffing

Skills

Languages: Python, TypeScript, JavaScript (Node.js), PostgreSQL, NoSQL, ARM Assembly/C Lang.

Platforms & Systems: AWS, ARM, Docker, Git, Kubernetes.

Libraries & Frameworks: React, Jest, MaterialUI, FastAPI, Boto3, Flask.

Accomplishments

CPL PV AI Hackathon: Winner of AI Innovation Award for Consistent Character Generation Using GenAI. 2024

PV Network Edge Hackathon: Winner for developing an AWS cost calculator for deployments. 2022

Armathon: Winner of the UNICEF Sustainability Challenge to end world hunger. 2021